

Joanna Zou

Email: jjzou@mit.edu
Website: joannajzou@github.io

Phone: (573) 529-2971
LinkedIn: @joannazou

Bio

Applied statistician with a background in engineering physics, training in the mathematics of machine learning, and experience in bringing theory into practice in materials science, renewable energy, natural hazard mitigation, structural analysis, signal processing, and remote sensing.

Interests: Bayesian inference, machine learning (ML), stochastic processes, generative diffusion models, sampling algorithms, uncertainty quantification, model-data fusion, data science

Skills: Python, Julia, HPC, Tensorflow, Matlab, LAMMPS

Languages: English (native), Chinese (fluent), German (basic), French (basic)

Education

PhD in Computational Science & Engineering

Massachusetts Institute of Technology
Cambridge, MA 2022 - 2025

MS in Structural Engineering

Stanford University
Palo Alto, CA 2018 - 2020

BS in Civil Engineering

Columbia University
GPA: 3.95/4.00 (*magna cum laude*)
New York, NY 2014 - 2018

Select Talks and Papers

- [C] "Goal-oriented learning of ergodic diffusions using error bounds on path-based observables." *SIAM DS*. 2025.
- [C] "Active learning of energy-based models via stochastic Stein variational gradient descent." *SIAM UQ*. 2024
- [C] "Adaptive importance sampling for gradient-based dimension reduction in stochastic systems." *UNCECOMP*. 2023.
- [C] "Bayesian modeling and active learning for molecular dynamics: fitting ML interatomic potentials to data." *CECAM Workshop*. 2022.
- [J] "Virtual sensing of subsoil strain response in monopile-based offshore wind turbines via Gaussian process latent force models." **J. Zou**, E. Lourens, A. Ciciello. *MSSP*, 200. 2023.
- [J] "Surrogate modeling for the seismic response estimation of residential wood frame structures." **J. Zou**, D. Welch, A. Zsarnoczay, A. Tafiandis, G. Deierlein. *WCEE Proceedings*. 2020.

Experience

Massachusetts Institute of Technology

PhD Student Researcher

Cambridge, MA
Feb. 2022 - Present

Research experience:

- accelerate ML training and sampling of informative data using a novel kernelized active learning algorithm based on principles of gradient flow
- curate efficient training datasets using a diversity-based subsampling algorithm based on determinantal point processes
- define a cost-effective dimension reduction algorithm for diffusion models via a multifidelity average gradient outer product (AGOP) of model outputs

Other experience:

- develop custom software packages in Julia for performant and scalable Bayesian inference of ML models for molecular materials science
- serve as a peer reviewer for academic journals (DCE, MSSP)
- organize a SIAM conf. minisymposium and interlab journal clubs
- serve in leadership of 2 student organizations and mentor prosp. PhD applicants

University of Potsdam Dept. of Mathematics

PhD Research Fellow

Potsdam, Germany
Jun. 2024 - Aug. 2024

- define machine learning objectives based on variational error bounds which improve sample efficiency and fidelity of pathwise properties of diffusion models
- derive proof of the bound and an exact computable gradient for optimization

Sandia National Laboratories

Graduate Research Intern

Livermore, CA
Jun. 2023 - Aug. 2023

- reduce cost of uncertainty quantification of diffusion models using an adaptive importance sampling-based control variate framework for dimension reduction
- validate method on Monte Carlo experiments with LAMMPS molecular simulation

Delft University of Technology

Fulbright US Research Fellow

Delft, Netherlands
Jan. 2021 - Dec. 2021

- develop and validate a latent variable model to predict strain-induced damage in offshore wind turbines using sparse acceleration data and Bayesian filtering
- use signal processing on 2-year vibration dataset to select validation samples

NHERI-SimCenter

Software Development & Research Associate

Palo Alto, CA
Aug. 2020 - Dec. 2020

- write Python code to develop natural hazards simulation software utilizing machine learning and HPC computing for large-scale infrastructure risk analysis
- establish parallel computing workflow for regional-level earthquake simulation

Stanford University

Masters Student Researcher

Palo Alto, CA
Aug. 2018 - May 2020

Research experience:

- accelerate probabilistic risk analysis by developing heteroscedastic Gaussian process surrogate models of the structural response of buildings to seismic load
- generate data using historic seismic recordings in numerical dynamic simulations
- train and tune a U-Net CNN for land use classification on satellite radar imagery
- curate features of footstep vibration data for a SVM classifier of gait anomalies

Other experience:

- create and deliver a 2-part lecture on convex optimization for a graduate course
- tutor students in Bachelor/Master courses in Math, Statistics, Computer Science

Arup

Advanced Technology & Research Intern

San Francisco, CA
Jun. 2018 - Aug. 2018

- develop probabilistic seismic risk assessment of building portfolios
- create finite element models in LS-DYNA for soil-structure interaction analysis